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Student Number

2014

HSC MID COURSE
EXAMINATION

Extension 1 Mathematics

11th March 2014

General Instructions

- Reading time – 5 minutes
- Working time 2 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 70

Section I - Pages 2 - 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages 7 - 10

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
Total	/70

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 25% of the Higher School Certificate Course Assessment.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10

1 Consider the polynomial $P(x) = 2x^5 - 3x - 7$. Which statement is most correct:

- (A) $P(x)$ is monic, of degree 5 with a constant term -7
- (B) $P(x)$ has a leading coefficient 2, is of degree 5 with a constant term -7
- (C) $P(x)$ has a leading coefficient 2, is of degree 5 with a constant term 7
- (D) $P(x)$ has a leading coefficient 2, is of degree 3 with a constant term -7

2 If $(x - 2)$ is a factor of the polynomial $P(x) = x^3 + 2x^2 - 5x + k$ then

- (A) $k = 10$
- (B) $k = -10$
- (C) $k = 6$
- (D) $k = -6$

3 Which integral is obtained when the substitution $u = 2 - x$ is applied to

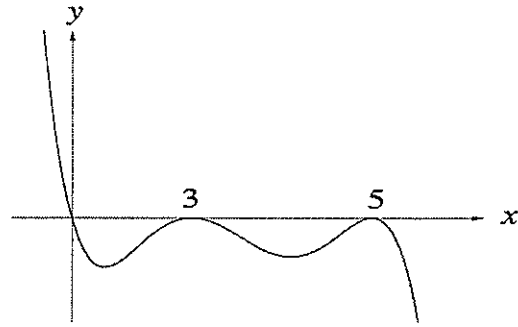
$$\int x\sqrt{2-x} dx$$

- (A) $\int (2-u)\sqrt{2-u} du$
- (B) $\int (2-u)\sqrt{u} du$
- (C) $\int (u-2)\sqrt{u} du$
- (D) $-\int (u-2)\sqrt{u} du$

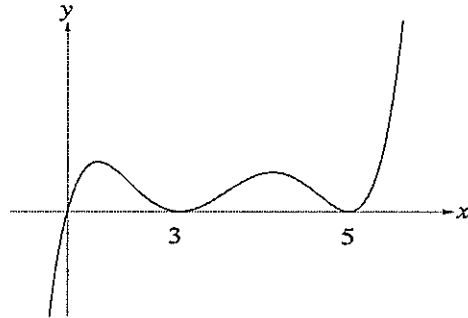
4

Which diagram best represents the graph $y = x(3-x)^3(x-5)^2$

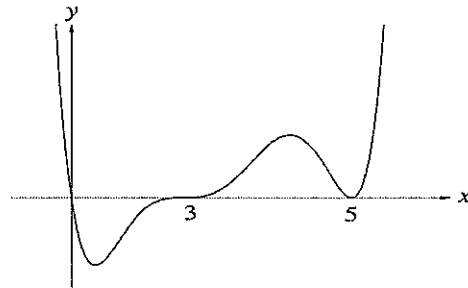
(A)



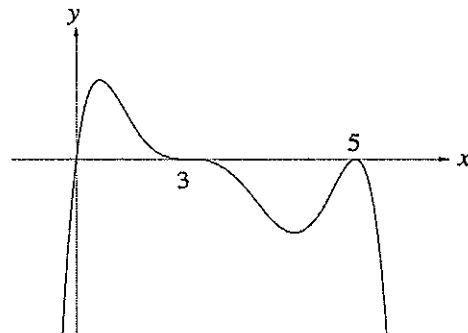
(B)



(C)



(D)



5 Let $|a| \leq 1$. What is the general solution of $\cos 3x = a$?

(A) $x = 2n\pi \pm \frac{\cos^{-1} a}{3}$

(B) $x = \frac{2n\pi \pm \cos^{-1} a}{3}$

(C) $x = n\pi \pm \frac{\cos^{-1} a}{3}$

(D) $x = \frac{n\pi \pm \cos^{-1} a}{3}$

6 Which of the following best describes the curve whose parametric equations are:

$$x = \frac{1}{2}\cos\theta, \quad y = \frac{1}{2}\sin\theta, \quad 0 \leq \theta \leq \pi$$

(A) A circle of radius $\frac{1}{2}$

(B) A circle of radius 4

(C) A semi-circle of radius $\frac{1}{2}$

(D) A semi- circle of radius 4

7 The coordinates of the point P that divides the interval joining $(-1, -6)$ and $(4, -1)$ internally in the ratio 3 : 2 are

(A) $P\left(-\frac{11}{5}, -\frac{16}{5}\right)$

(B) $P(1, -4)$

(C) $P(2, -3)$

(D) $P\left(-\frac{14}{5}, -\frac{16}{5}\right)$

8

The exact value of $\cos\left(\frac{7\pi}{12}\right)$

(A) $\frac{\sqrt{2}-\sqrt{6}}{4}$

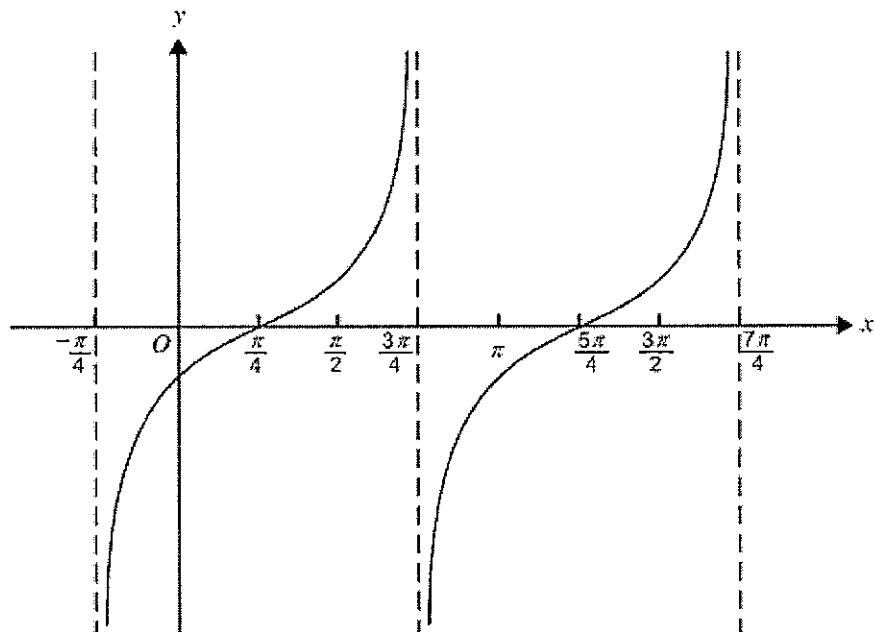
(B) $\frac{\sqrt{2}+1}{2}$

(C) $\frac{\sqrt{2}+\sqrt{6}}{4}$

(D) $\sqrt{6}$

9

The diagram shows two cycles of the graph of a circular function



The period of the circular function is

(A) $\frac{\pi}{2}$

(B) $\frac{3\pi}{4}$

(C) π

(D) 2π

10

The exact volume when the function $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{2}$ is rotated around the x -axis is:

(A) $\frac{\pi^2}{4}$

(B) 1

(C) π

(D) $\frac{\pi^2}{2}$

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11(15 marks) Use a SEPARATE writing booklet

- (a) The polynomial $P(x) = x^3 + x^2 - kx - 4$ has roots α, β, γ .
- (i) Find the value of $\alpha + \beta + \gamma$ 1
- (ii) Find the value of $\alpha\beta\gamma$ 1
- (iii) Find an expression for $\alpha^2 + \beta^2 + \gamma^2$ 2
- (iv) It is known that two of the roots are equal in magnitude but opposite in sign 2
- Find the third root and hence find the value of k .
- (b) Integrate $\int 2 \cos^2 x \, dx$ 2
- (c) Prove that 2
- $$\frac{\cos 2\theta}{\cos \theta} + \frac{\sin 2\theta}{\sin \theta} = \frac{4 \cos^2 \theta - 1}{\cos \theta}$$
- (d) (i) Find the vertical and horizontal asymptotes of the hyperbola $y = \frac{2x-1}{x-3}$ 3
- and hence sketch the graph of $y = \frac{2x-1}{x-3}$
- (ii) Hence, or otherwise, find the values of x for which $\frac{2x-1}{x-3} \leq 4$ 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Integrate $\int \frac{x^2 dx}{\sqrt{x^3 + 3}}$ using the substitution $u = x^3 + 3$ 2
- (b) Evaluate $\int_1^{25} \frac{1}{\sqrt{x}\sqrt{1+\sqrt{x}}} dx$ using the substitution $u = 1 + \sqrt{x}$ leaving your solution as an exact value. 3
- (c) Evaluate $\int_0^{\frac{\pi}{2}} \sin^3 x \cos x dx$ 3
- (d) Divide $P(x) = x^4 - 3x^3 + 5x^2 - 8x - 5$ by $x - 3$ 3
Express the result in the form $P(x) = A(x)Q(x) + R(x)$
- (e) Use the substitution $t = \tan \frac{x}{2}$ to solve the equation $5\sin x - 5\cos x + 1 = 0$ 4
for $0 \leq x \leq 360^\circ$, correct to two decimal places.

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) Show that $f(x) = x^3 - 2x^2 - x + 1$ has a root between $x = 0$ and $x = 1$ and use the bisection method (halving the interval) twice to determine an estimate of the root. 3
- (b) Use a first approximation of $x = 1.4$ with one application of Newton's method to solve $3\sin x - 2x = 0$ correct to 3 decimal places. 3
- (c) (i) On the same set of axes sketch 2
 $y = \sin^2 x$ and $y = \cos^2 x$ for $0 \leq x \leq 2\pi$
- (ii) Show that $y = \sin^2 x$ and $y = \cos^2 x$ intersect at $x = \frac{\pi}{4}$ 2
- (iii) Find the exact area bounded by the graphs $y = \cos^2 x$, $y = \sin^2 x$ 3
 $x = 0$ and $x = \frac{\pi}{4}$.
- (d) A polynomial $P(x)$ of degree 3 has zeros at -1, 2 and 3. Its leading coefficient is 4. Write down the polynomial. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) For the curve $y = 5xe^x$
- (i) Determine the stationary point and the point of inflexion 3
 - (ii) Sketch the curve 2
 - (iii) From your sketch, show that the equation $5xe^x = k$ has:
 - (α) Two roots if $-\frac{5}{e} < k < 0$ 1
 - (β) One root if $k > 0$ 1
- (b) (i) Express $\sin 4t + \sqrt{3} \cos 4t$ in the form $R \sin(4t + \alpha)$ where α is in radians 3
- (ii) Hence solve $\sin 4t + \sqrt{3} \cos 4t = 0$ for $0 \leq t \leq \pi$ 2
- (c) Let $P(x) = (x+2)(x-4)Q(x) + a(x+2) + b$, where $Q(x)$ is a polynomial and a and b are real numbers. 3
- When $P(x)$ is divided by $(x+2)$ the remainder is -5
- When $P(x)$ is divided by $(x-4)$ the remainder is 7
- What is the remainder when the polynomial is divided by $(x+2)(x-4)$?

End of Paper



Killara
HIGH SCHOOL

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Student Number

Multiple Choice Answer Sheet (Please detach)

Completely fill the response oval representing the most correct answer.

1. A ○ B ○ C ○ D ○
2. A ○ B ○ C ○ D ○
3. A ○ B ○ C ○ D ○
4. A ○ B ○ C ○ D ○
5. A ○ B ○ C ○ D ○
6. A ○ B ○ C ○ D ○
7. A ○ B ○ C ○ D ○
8. A ○ B ○ C ○ D ○
9. A ○ B ○ C ○ D ○
10. A ○ B ○ C ○ D ○

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

1. $\boxed{B}$

2. $P(2)=0. \quad 6+k=0 \quad k=-6. \quad \boxed{D}$

3. $\int x \sqrt{2-x} dx.$

$u = 2-x. \Rightarrow x = 2-u$

$du = -dx.$

$\therefore -\int (2-u) \sqrt{u} du. = \int (u-2) \sqrt{u} du. \quad \boxed{C}$

4. D

5. $B.$

$$6. \quad \left. \begin{array}{l} x = \frac{1}{2} \cos \theta. \\ y = \frac{1}{2} \sin \theta. \end{array} \right\} \begin{array}{l} \cos \theta = 2x \\ \sin \theta = 2y \end{array} \quad 0 \leq \theta \leq \pi$$

$$\therefore x^2 + y^2 = \frac{1}{4}.$$

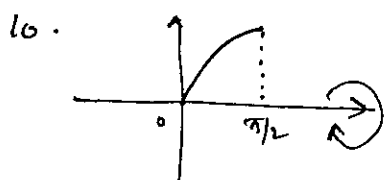
$\boxed{C}$

7. $(-1, -6) \quad (4, -1).$
3:2

$$P \left[\frac{3 \times 4 + 2 \times -1}{3+2}, \frac{3 \times -1 + 2 \times 6}{3+2} \right] = (2, -3) \quad \boxed{C}$$

8. $\cos \frac{7\pi}{12}$  negative answer. $\frac{\sqrt{2}-\sqrt{6}}{4}$ $\boxed{A}$

9. C



$$\pi \int_0^{\pi/2} \sin^2 x dx = \frac{\pi}{2} \int_0^{\pi/2} 1 - \cos 2x dx$$

$$= \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2} = \frac{\pi}{2} \left[\frac{\pi}{2} - 0 - 0 + 0 \right] = \frac{\pi^2}{4} \quad \boxed{A}$$

Question 11.

$$a) \cdot p(x) = x^3 + x^2 - kx - 4.$$

$$(i) \quad \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{1}{1} = -1$$

$$(ii) \quad \alpha\beta\gamma = -\frac{d}{a} = 4.$$

$$\begin{aligned} (iii) \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\ &= 1 - 2(-k) \\ &= 1 + 2k. \end{aligned}$$

$$(iv) \quad \alpha, -\alpha, \beta.$$

$$\alpha + -\alpha + \beta = -1 \quad \therefore \beta = -1.$$

$$\begin{aligned} p(-1) &= 0. \quad (-1)^3 + (-1)^2 - k(-1) - 4 = 0. \\ -4 + k &= 0 \\ \therefore k &= 4. \end{aligned}$$

$$\begin{aligned} b) \cdot \int 2 \cos^2 x \, dx \\ &= 2 \int \frac{1 + \cos 2x}{2} \, dx. \\ &= x + \frac{\sin 2x}{2} + C. \end{aligned}$$

$$\begin{aligned} c) \quad &\frac{\cos 2\theta}{\cos \theta} + \frac{\sin 2\theta}{\sin \theta} \\ &\frac{(2\cos^2 \theta - 1) \sin \theta + 2 \sin \theta \cos \theta \cdot \cos \theta}{\sin \theta \cdot \cos \theta} \\ &= \frac{2\cos^2 \theta - 1 + 2\cos^2 \theta}{\cos \theta} \\ &= \frac{4\cos^2 \theta - 1}{\cos \theta}. \end{aligned}$$

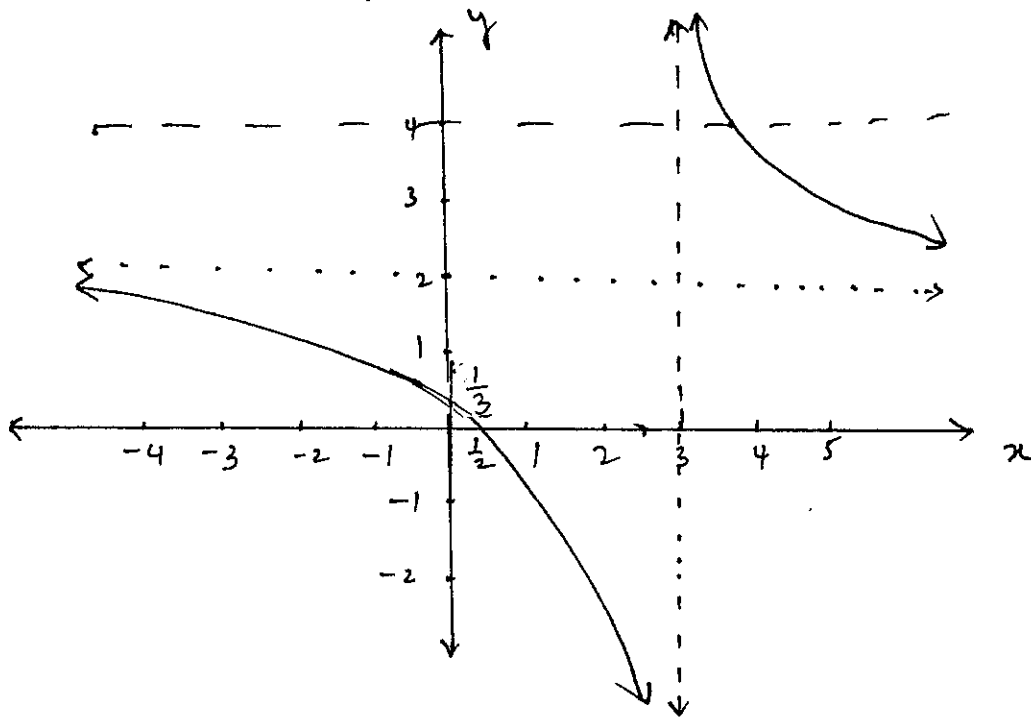
d)

$$(1) \quad y = \frac{2x-1}{x-3} = \frac{2(x-3) + 5}{x-3}$$

$$= 2 + \frac{5}{x-3}$$

Vertical asymptote $x=3$.

Horizontal asymptote $y=2$.



(11) $\frac{2x-1}{x-3} = 4$.

$$2x-1 = 4x-12$$

$$2x = 11$$

$$x = 5.5$$

$$x > 5.5, \quad x < 3$$

Question 12

4

$$\begin{aligned}
 a) \quad & \int \frac{x^2 dx}{\sqrt{x^3+3}} \quad \left| \quad \begin{aligned} & \frac{1}{3} \int \frac{3x^2 dx}{\sqrt{x^3+3}} \\ & = \frac{1}{3} \int \frac{du}{\sqrt{u}} \\ & \frac{1}{3} \int u^{-1/2} du \\ & \frac{1}{3} \cdot 2 u^{1/2} + C = \frac{2}{3} \sqrt{x^3+3} + C \end{aligned} \right.
 \end{aligned}$$

$$b) \quad \int_1^{25} \frac{1}{\sqrt{x}(\sqrt{1+\sqrt{x}})} dx.$$

$$\begin{aligned}
 u = 1 + \sqrt{x} \quad & \left| \quad \begin{aligned} x=1 \quad u=2 \\ x=25 \quad u=6 \end{aligned} \right. \\
 du = \frac{1}{2\sqrt{x}} dx
 \end{aligned}$$

$$\int_2^6 \frac{2 du}{u}$$

$$= 2 [\ln u]_2^6 = 2 \ln \frac{6}{2} = \underline{\underline{2 \ln 3}}$$

$$c) \quad \int_0^{\pi/2} \sin^3 x \cos x dx$$

$$\left[\frac{\sin^4 x}{4} \right]_0^{\pi/2} = \frac{1}{4} [1 - 0] = \underline{\underline{\frac{1}{4}}}$$

$$\begin{array}{r}
 d) \quad x-3 \overline{) \begin{array}{r} x^3 + 5x + 7 \\ x^4 - 3x^3 + 5x^2 - 8x - 5 \\ \underline{x^4 - 3x^2} \\ 0 + 5x^2 - 8x \\ \underline{5x^2 - 15x} \\ +7x - 5 \\ \underline{7x - 21} \\ 16 \end{array}}
 \end{array}$$

$$\therefore x^4 - 3x^3 + 5x^2 - 8x - 5 \equiv (x^2 + 5x + 7)(x - 3) + 16.$$

$$e) \quad t = \tan \frac{x}{2}$$

$$5 \sin x - 5 \cos x + 1 = 0.$$

$$5 \times \frac{2t}{1+t^2} - 5 \frac{1-t^2}{1+t^2} + 1 = 0.$$

$$10t - 5 + 5t^2 + 1 + t^2 = 0.$$

$$6t^2 + 10t - 4 = 0.$$

$$3t^2 + 5t - 2 = 0.$$

$$(t+2)(3t-1) = 0.$$

$$t = -2$$

$$t = \frac{1}{3}$$

$$0 \leq x \leq 360^\circ$$

$$0 \leq \frac{x}{2} \leq 180^\circ$$

$$\frac{x}{2} = 116.57^\circ$$

$$\frac{x}{2} = \tan^{-1}\left(\frac{1}{3}\right) = 18.43^\circ \dots$$

$$x = 36.86^\circ$$

$$\therefore x = 233.13^\circ.$$

Test $x = 180^\circ$

$$5 \sin 180 - 5 \cos 180 + 1 = 6 \neq 0.$$

Question 13

$$a) \quad f(x) = x^3 - 2x^2 - x + 1$$

$$f(0) = 1 \quad f(1) = -1$$

$f(x)$ is continuous and change of sign between $f(0)$ and $f(1)$. $\therefore$ Solution exists between 0 and 1.

(6)

Estimate 1.

$$x_1 = \frac{0+1}{2} = \frac{1}{2}$$

$$f\left(\frac{1}{2}\right) = 0.125 \quad ; \quad f(1) = -1.$$

Estimate 2

$$x_2 = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4}$$

$$b) \quad x_0 = 1.4.$$

$$f(x) = 3\sin x - 2x$$

$$f'(x) = 3\cos x - 2$$

$$x_1 = x_0 - \frac{f(1.4)}{f'(1.4)} = 1.5049\dots$$

$$= \underline{\underline{1.505}}.$$

c)

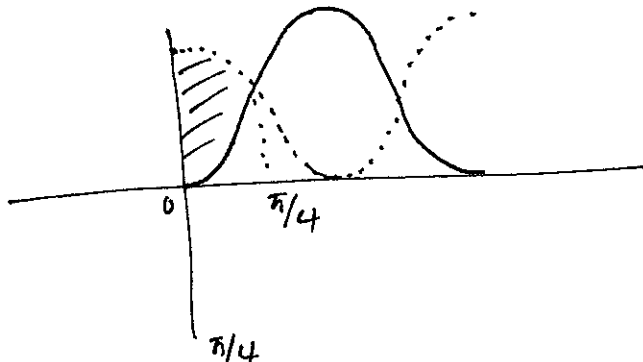
$$(ii) \quad x = \pi/4$$

$$y = \sin^2 \frac{\pi}{4} = \frac{1}{2}$$

$$y = \cos^2 \frac{\pi}{4} = \frac{1}{2}$$

$x = \frac{\pi}{4}$ satisfies both equations.

(iii)



$$\text{Area} = \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx$$

$$= \int_0^{\pi/4} \cos 2x \, dx$$

$$= \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} = \frac{1}{2} [1 - 0] = \frac{1}{2} \underline{\underline{\text{unit}^2}}$$

$$d) \quad P(x) = 4(x+1)(x-2)(x-3).$$

Question 14

a) $y = 5xe^x$

(i) For stationary point $\frac{dy}{dx} = 0$.

$$5[xe^x + e^x] = 0.$$

$$5e^x[x+1] = 0 \quad e^x > 0 \text{ for all } x.$$

$$\therefore x = -1$$

$$x = -1, \quad y = -5e^{-1} = -\frac{5}{e} = -1.83 \quad \left(-1, -\frac{5}{e}\right).$$

Point of inflection $\frac{d^2y}{dx^2} = 0$.

$$5e^x \cdot 1 + 5e^x(x+1) = 0.$$

$$5e^x[1+x+1] = 0 \quad e^x > 0 \text{ for all } x$$

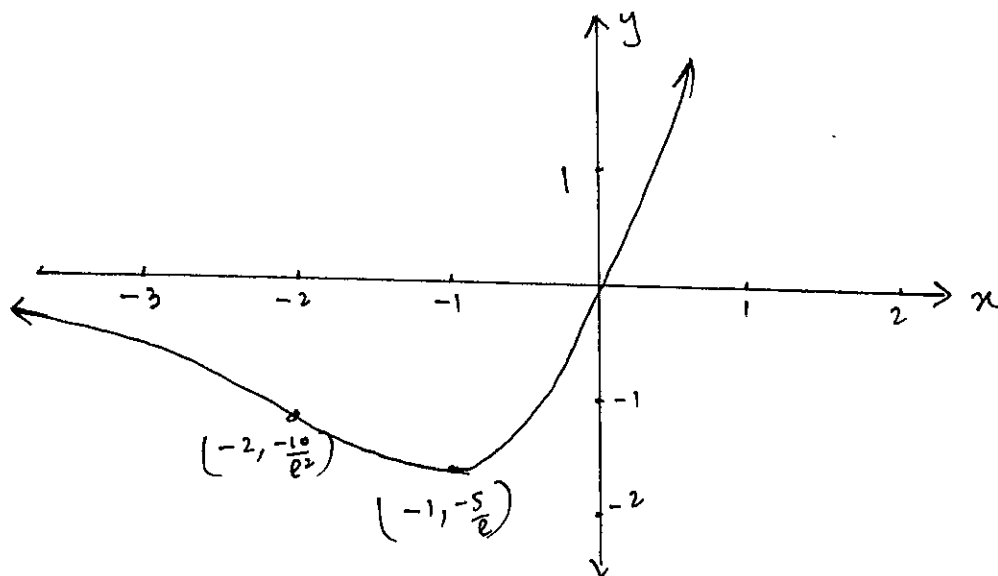
$$\therefore x = -2$$

$$x = -2, \quad y = -10e^{-2} = -\frac{10}{e^2} \approx -1.35$$

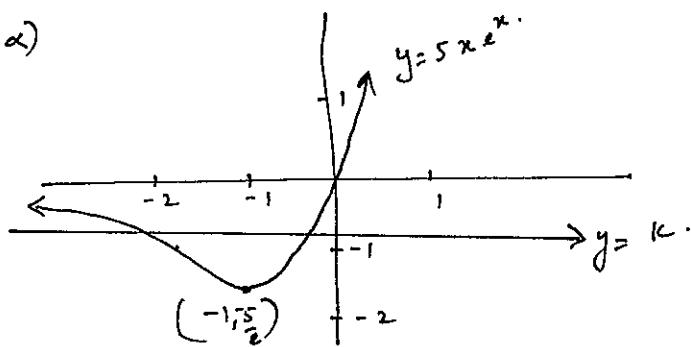
x	-3	-2	-1
$\frac{d^2y}{dx^2}$	-0.25	0	1.83.

Change in Concavity
 $\therefore$ Point of inflection at $\left(-2, -\frac{10}{e^2}\right)$

(ii) $x \rightarrow -\infty, y \rightarrow 0$; $x \rightarrow \infty, y = \infty$



(iii) α)



⑨

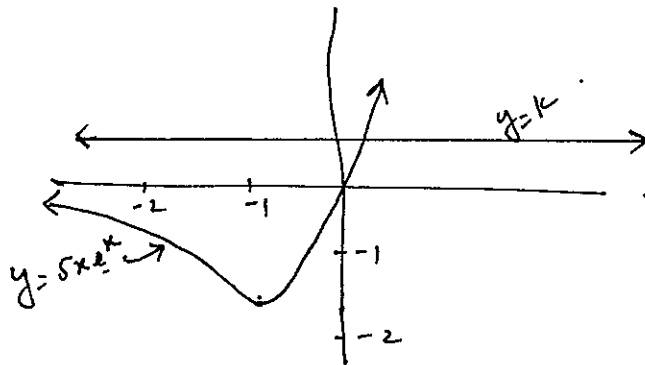
For $-\frac{5}{2} < k < 0$, the line

$y = k$ cuts the curve

$y = 5xe^x$ at two points.

Hence $5xe^x = k$ has two solutions in this region.

β)



for $k > 0$, the line $y = k$ cuts $y = 5xe^x$ only in one point.

∴ only one solution.

b)

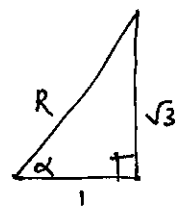
(1)

$$\sin 4t + \sqrt{3} \cos 4t \equiv R \sin(4t + \alpha)$$

$$R[\sin 4t \cos \alpha + \cos 4t \sin \alpha]$$

$$\cos \alpha = \frac{1}{R} \quad \sin \alpha = \frac{\sqrt{3}}{R}$$

$$R = \sqrt{2} \quad \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$$



$$\therefore 2 \sin\left(4t + \frac{\pi}{3}\right)$$

$$0 \leq t \leq \pi$$

$$0 \leq 4t \leq 4\pi$$

$$(11) \quad \sin 4t + \sqrt{3} \cos 4t = 0$$

$$2 \sin\left(4t + \frac{\pi}{3}\right) = 0$$

$$4t + \frac{\pi}{3} = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$t = \frac{\pi}{12}, \frac{\pi}{6}, \frac{5\pi}{12}, \frac{2\pi}{3}, \frac{11\pi}{12}$$

$$\left. \begin{array}{l} t = \frac{\pi - \frac{\pi}{3}}{4} \end{array} \right\}$$

(12)

$$c) \quad p(x) = (x+2)(x-4)q(x) + a(x+2) + b.$$

$$p(-2) = -5 \quad \therefore b = -5$$

$$p(4) = 7 \quad \therefore 6a - 5 = 7$$

$$6a = 12$$

$$\therefore a = 2$$

$$\therefore \text{Remainder is } a(x+2) + b$$

$$= 2(x+2) - 5$$

$$= \underline{\underline{2x - 1}}$$